Transition Systems and Linear Temporal Logic

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Linear Temporal Logic (LTL)

Introduction to LTL

- LTL can be used to specify important system properties.
- For reactive systems, correctness depends on the *executions* of the system – not only the input and output of a computation – and on fairness issues.
    - Temporal logic is a formalism for treating these aspects.
- LTL extends propositional or predicate logic by modalities that permit to referral to the infinite behavior of a reactive system.

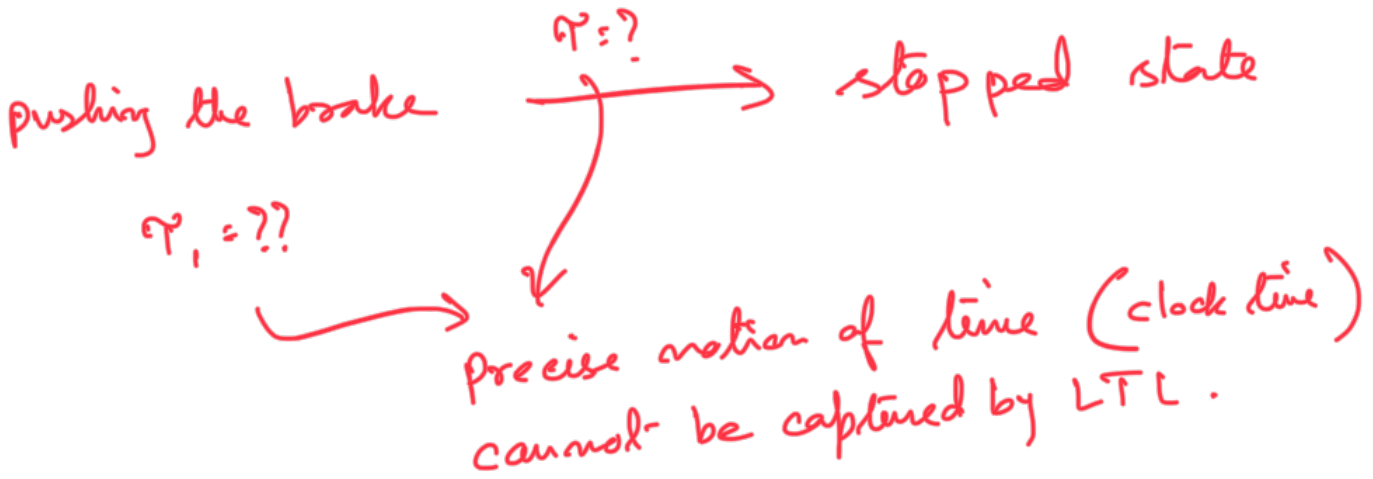
Introduction to LTL

- Provides a very intuitive but mathematically precise notation for expressing properties about the relation between the state labels in execution.
- Imposes temporal modalities on top of predicate logic.
- Although the word "temporal" suggests a relationship with the real-time behavior of a reactive system, this is only true in an abstract sense
    - Allows for the specification of the relative order of events

Introduction to LTL

Example:

The car stops once the car pushes the brake.



Introduction to LTL

Syntax of LTL

Definition 5.1. Syntax of LTL

LTL formulae over the set AP of atomic proposition are formed according to the following grammar:[1]

$$\varphi ::= \text{true} \mid a \mid \varphi_1 \wedge \varphi_2 \mid \neg\varphi \mid \bigcirc \varphi \mid \varphi_1 \mathsf{U} \varphi_2$$

where $a \in AP$. ■

Syntax of LTL

Syntax of LTL

$$
\begin{array}{lcl}
\varphi_1 \vee \varphi_2 & \stackrel{\text{def}}{=} & \neg(\neg\varphi_1 \wedge \neg\varphi_2) \\
\varphi_1 \rightarrow \varphi_2 & \stackrel{\text{def}}{=} & \neg\varphi_1 \ \vee \ \varphi_2 \\
\varphi_1 \leftrightarrow \varphi_2 & \stackrel{\text{def}}{=} & (\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1) \\
\varphi_1 \oplus \varphi_2 & \stackrel{\text{def}}{=} & (\varphi_1 \wedge \neg\varphi_2) \ \vee \ (\varphi_2 \wedge \neg\varphi_1)
\end{array}
$$

.

Syntax of LTL



Figure 5.1: Intuitive semantics of temporal modalities.

Syntax of LTL

$\Box\Diamond a$: infinitely often

$\Diamond\Box a$: from some moment j onwards, only a-state will be true.

Examples of LTL

Mutual Exclusion problem

P_1 and P_2 : concurrent processes

i) non-crit ii) waiting iii) critical

wait$_i$ crit$_i$

Safety property: $crit_1 \cup crit_2$ ✗

$\Box (\neg crit_1 \vee \neg crit_2)$

the processes will try to enter the crit section perpetually (again and again)

Examples of LTL

Mutual exclusion problem

- Liveness property: Each process P_i is infinitely often in its crit sec.
  (progress)

$$\Box\Diamond crit_1 \wedge \Box\Diamond crit_2$$

?? Semaphores.

- Bounded Waiting (starvation-free property)

Every waiting P_i will eventually enter its crit sec.

$$\left(\Box\Diamond wait_1 \rightarrow \Box\Diamond crit_1\right) \wedge \left(\Box\Diamond wait_2 \rightarrow \Box\Diamond crit_2\right)$$

$$\Box\left(\Diamond wait_1 \rightarrow \Diamond crit_1\right)$$

Examples of LTL

A communication channel

- unidirectional
- capacity of the buffers is irrelevant.



Requirement:

whenever m is in the out-buffer of S, then m will eventually be received by the receiver buffer.

(i) $\Box\left(m \in S.out \longrightarrow \Diamond\,(m \in R.in)\right)$

$s_1\ s_2\ s_3\ \ldots$

$s_1 \models m \in S.out$, $s_2 \models m \notin S.out$, $s_2 \models m \notin R.in$

$s_3 \models m \in R.in$

$\Box\left(m \in S.out \rightarrow m \in S.out \ \mathcal{U}\ m \in R.in\right)$

resolves the problem

Examples of LTL

Sensor-based robotic tasks

Sensor propositions $X = \{x_1, \ldots, x_n\}$

System $Y = \{r_1, \ldots, r_m\}$

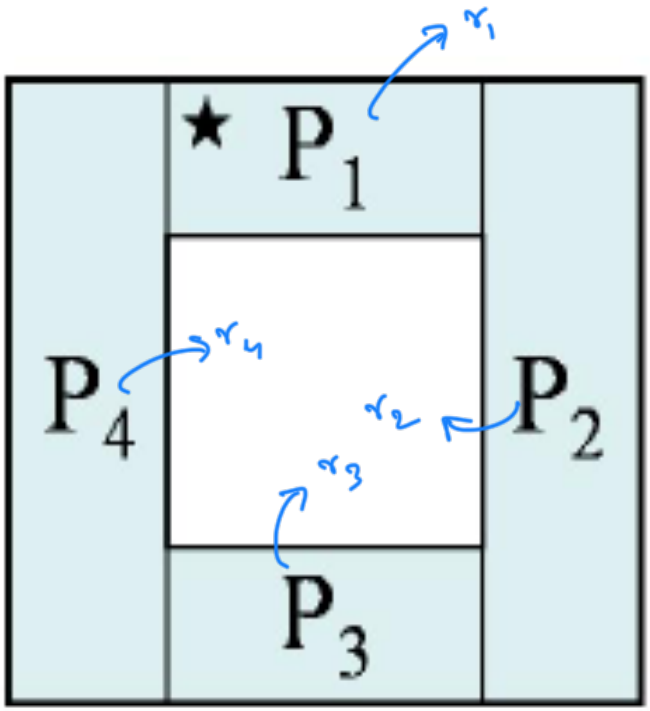
$\varphi = (\varphi_e \Rightarrow \varphi_s)$

↳ why?

$\varphi_e = \varphi_i^e \wedge \varphi_t^e \wedge \varphi_g^e$

$\varphi_s = \varphi_i^s \wedge \varphi_t^s \wedge \varphi_g^s$

↓ possible evolution of the system



Two entities

(i) System (robot)

(ii) Environment

$P_i \cap P_j = \phi$ for all i, j

[pair-wise disjoint]

Fig. 1: The workspace of Example 1. The initial position of the robot is marked with a star.

Sensor-based robotic tasks

Example 1: Consider a robot that is moving in the workspace shown in Fig. 1 consisting of four areas labelled $P_1, ..., P_4$ (which define the system propositions $\mathcal{Y} = \{r_1, \ldots, r_4\}$). Initially, the robot is placed somewhere in region P_1. In natural language, the desired specification for the robot is: *Look for Waldo in regions* P_2 *and* P_4*, if you find him, stay where you are, and if not, keep looking.*

Sensor-based robotic tasks

Environment properties

$\varphi_e = \varphi_i^e \wedge \varphi_t^e \wedge \varphi_g^e$

- $\varphi_i^e = (\neg s^{\text{Waldo}})$

Since Waldo is a part of the environment, we have one proposition $X = \{s^{\text{Waldo}}\}$

(1)

- $\varphi_t^e = \left\{ \begin{array}{l} \square((\neg r_2 \wedge \neg r_4) \Rightarrow (\bigcirc s^{\text{Waldo}} \Leftrightarrow s^{\text{Waldo}})) \\ \wedge \square(s^{\text{Waldo}} \Rightarrow \bigcirc s^{\text{Waldo}}) \end{array} \right.$

(2)

$\varphi_g^e = \square\Diamond$ true

* Waldo is not moving until it is found

Sensor-based robotic tasks

System properties

$\varphi_s = \varphi_i^s \wedge \varphi_t^s \wedge \varphi_g^s$

$\varphi_i^s = (r_1 \wedge \neg r_2 \wedge \neg r_3 \wedge \neg r_4)$

$$\varphi_t^s = \begin{cases} \bigwedge \Box(r_1 \Rightarrow (\bigcirc r_1 \vee \bigcirc r_2 \vee \bigcirc r_4)) \\ \bigwedge \Box(r_2 \Rightarrow (\bigcirc r_1 \vee \bigcirc r_2 \vee \bigcirc r_3)) \\ \bigwedge \Box(r_3 \Rightarrow (\bigcirc r_2 \vee \bigcirc r_3 \vee \bigcirc r_4)) \\ \bigwedge \Box(r_4 \Rightarrow (\bigcirc r_1 \vee \bigcirc r_3 \vee \bigcirc r_4)) \\ \bigwedge \Box((\bigcirc r_1 \wedge \neg \bigcirc r_2 \wedge \neg \bigcirc r_3 \wedge \neg \bigcirc r_4) \\ \quad \vee (\neg \bigcirc r_1 \wedge \bigcirc r_2 \wedge \neg \bigcirc r_3 \wedge \neg \bigcirc r_4) \\ \quad \vee (\neg \bigcirc r_1 \wedge \neg \bigcirc r_2 \wedge \bigcirc r_3 \wedge \neg \bigcirc r_4) \\ \quad \vee (\neg \bigcirc r_1 \wedge \neg \bigcirc r_2 \wedge \neg \bigcirc r_3 \wedge \bigcirc r_4)) \\ \bigwedge_{i \in \{2,4\}} \Box((r_i \wedge \bigcirc s^{\text{Waldo}}) \Rightarrow \bigcirc r_i) \end{cases}$$

When the robot senses Waldo it should remain in the same region

$\varphi_g^s = \Box\Diamond(r_2 \vee s^{\text{Waldo}}) \wedge \Box\Diamond(r_4 \vee s^{\text{Waldo}})$

Thank you